

Implementation of Vanilla Transformer for English-Hindi Machine Translation

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1 Introduction

The objective of this project was to implement the Vanilla Transformer architecture proposed in the paper “Attention Is All You Need”. The model was implemented completely from scratch using PyTorch without using the built-in Transformer module. The task chosen was English-to-Hindi machine translation.

2 Model Architecture

The implemented Transformer consists of the following components:

- Positional Encoding
- Scaled Dot Product Attention
- Multi-Head Attention
- Feed Forward Network
- Encoder
- Decoder
- Final Linear Layer

The encoder processes the English sentence while the decoder generates the Hindi translation autoregressively.

3 Dataset

The English-Hindi parallel corpus was used for training and evaluation. The dataset was preprocessed by tokenizing the sentences, constructing vocabularies for both languages, numericalizing the tokens, and padding all sequences to a fixed maximum length.

4 Training

The model was trained using the Adam optimizer and CrossEntropy loss. Teacher forcing was used during training. After training, greedy decoding was used during inference to generate translations.

5 Results

The performance of the model was evaluated using the BLEU score.

Metric	Value
BLEU Score	0.000873%
Training Epochs	15

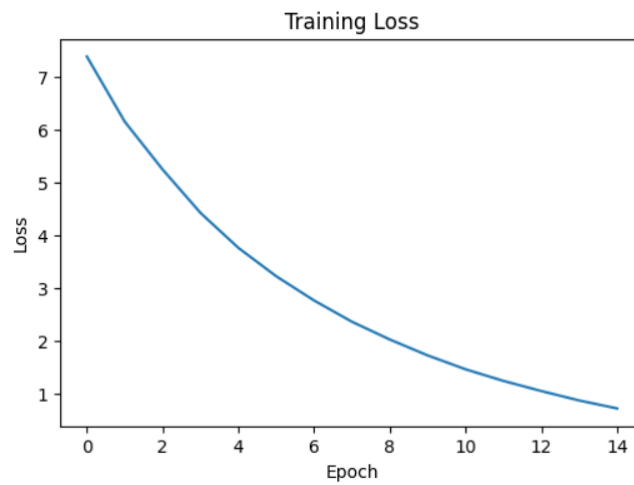


Figure 1: Training and Validation Loss